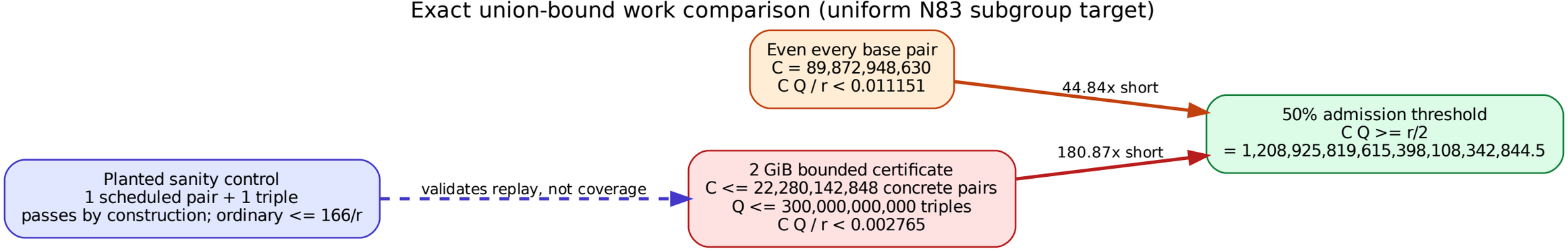
Exact union-bound work comparison (uniform N83 subgroup target)
Even every base pair
C = 89,872,948,630
C Q / r < 0.011151
44.84x short
50% admission threshold
C Q >= r/2
= 1,208,925,819,615,398,108,342,844.5
180.87x short
Planted sanity control
1 scheduled pair + 1 triple
passes by construction; ordinary <= 166/r
validates replay, not coverage
2 GiB bounded certificate
C <= 22,280,142,848 concrete pairs
Q <= 300,000,000,000 triples
C Q / r < 0.002765